COMPUTER SYSTEMS FUNDAMENTALS
(4COSC004W)

Week 1. Part 1 of 3

THE NATURE OF NUMBERS

Positional number systems

By the end of this lecture, you will:

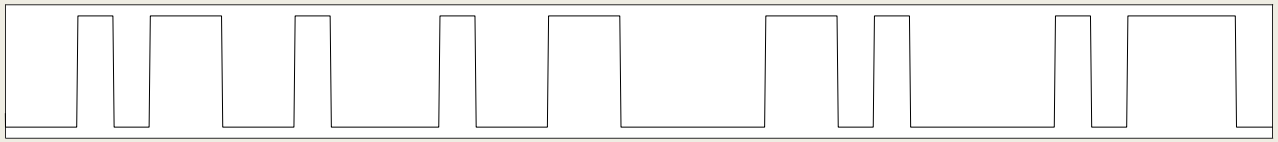
- Understand the concept of Positional Number Systems
- Be able to count and interpret natural numbers
  - *Positive (unsigned) Integers*
- Understand the following number systems:
  - *Decimal / Denary – Base 10*
  - *Binary - Base 2*
- Be able to count in Binary

Binary : Why use Binary?

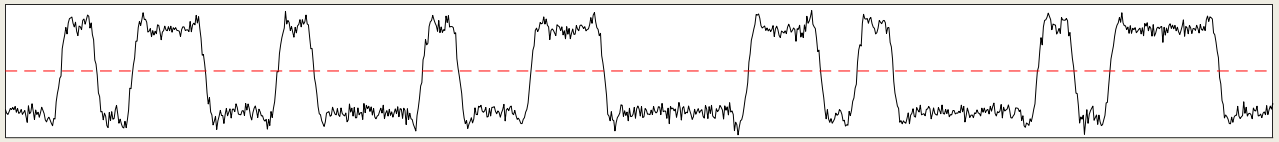
- Computer Systems use Binary to represent data (logic-0 and logic-1)
- Binary coding to code real-life signals



An example to an Analog Waveform



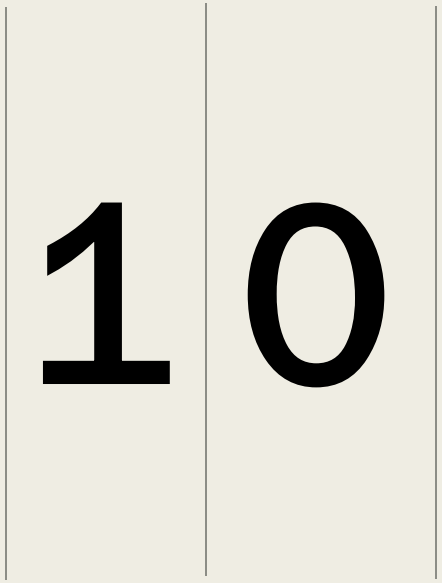
Binary coded Waveform to represent digital information.



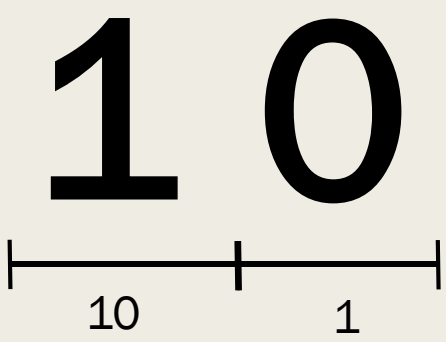
Binary coded Waveform with Noise. The code can still be acquired despite the distortion added.

- Binary is used for Digital systems
  - *On/Off*

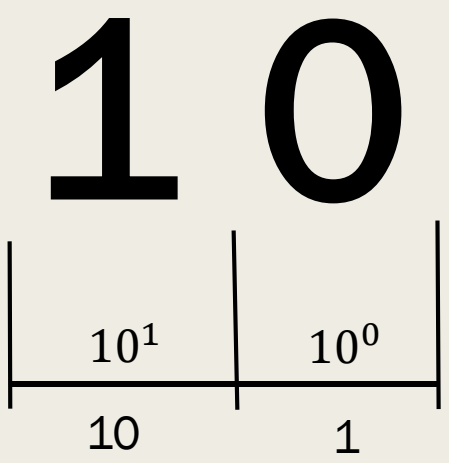
Decimal / Denary – Base 10



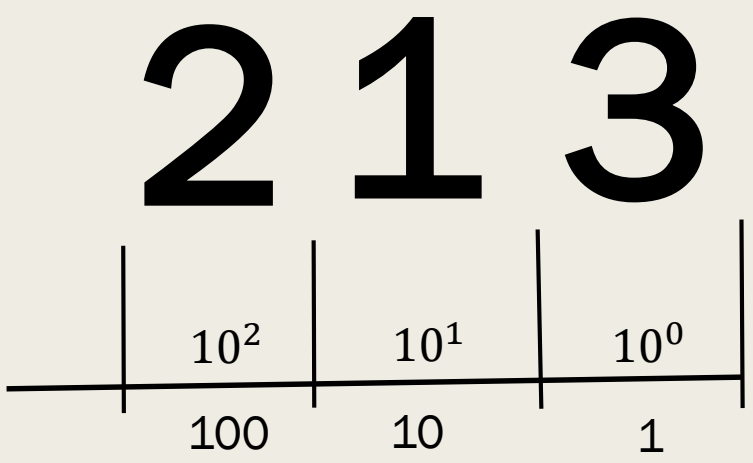
Decimal / Denary – Base 10



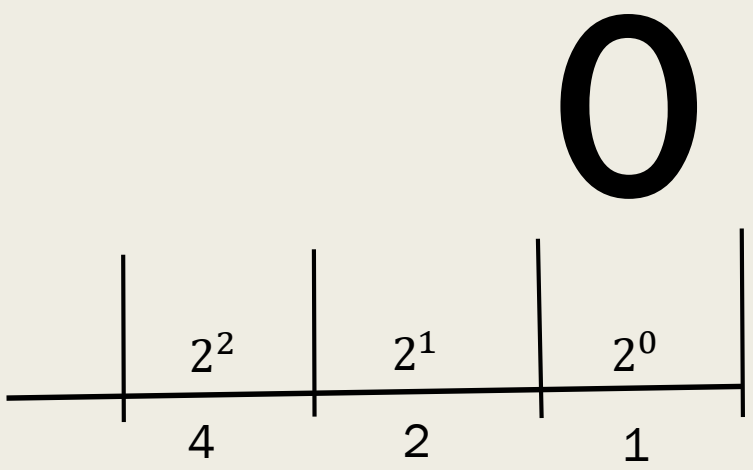
Decimal / Denary – Base 10



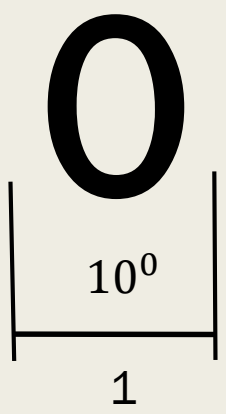
Decimal / Denary – Base 10



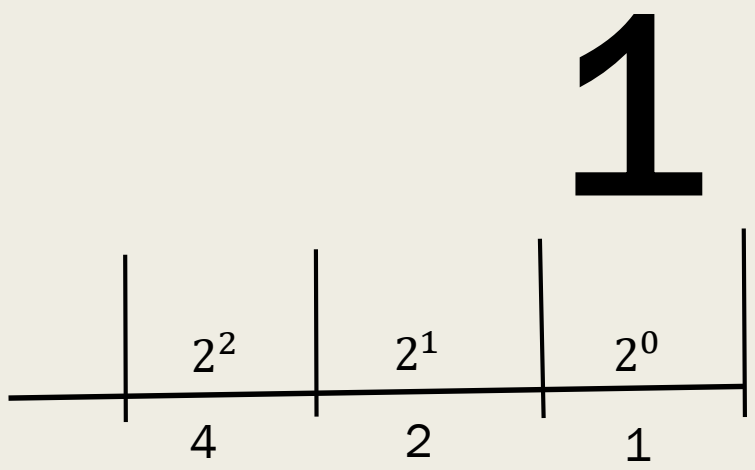
Base 2
Binary



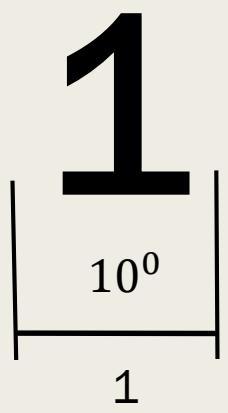
Base 10
Denary



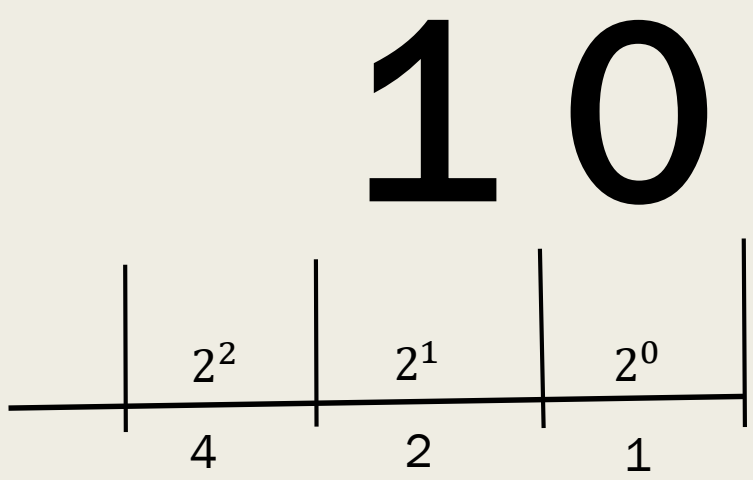
Base 2
Binary



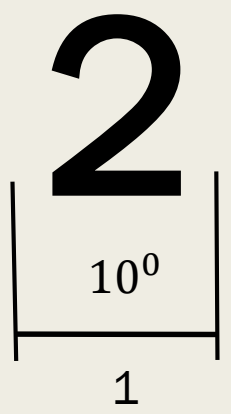
Base 10
Denary



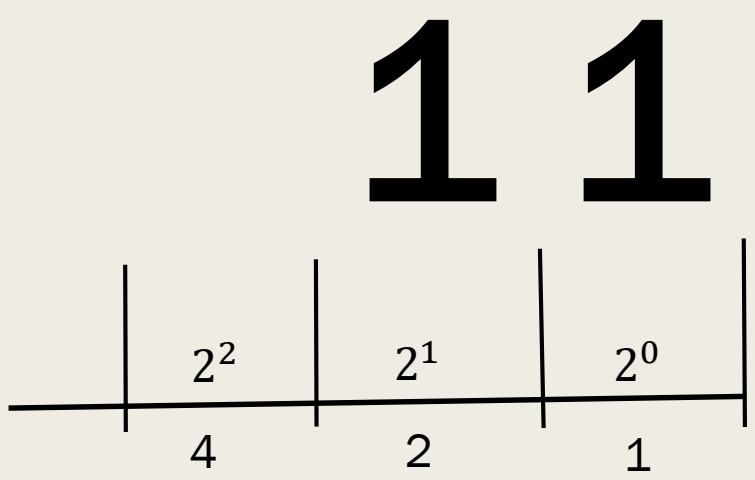
Base 2
Binary



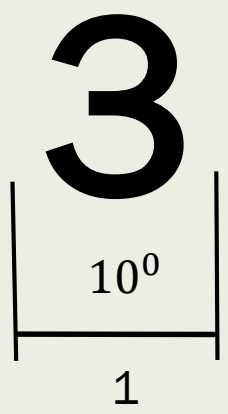
Base 10
Denary



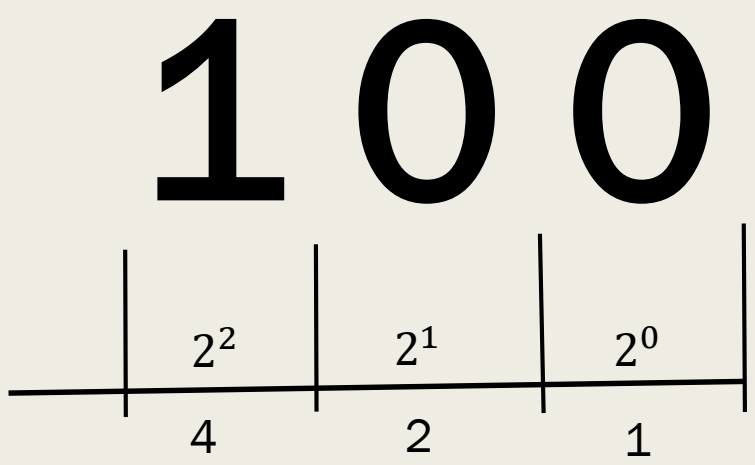
Base 2
Binary



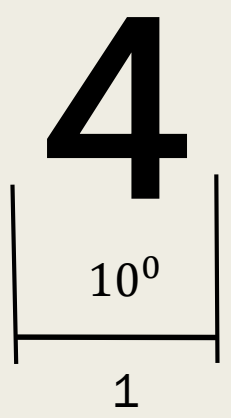
Base 10
Denary



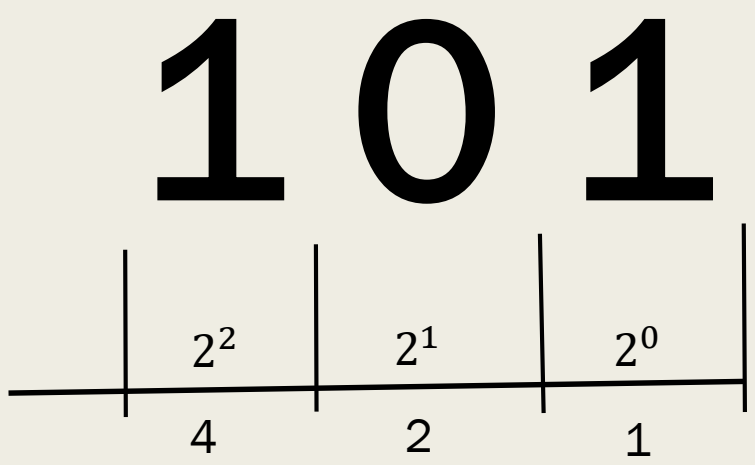
Base 2
Binary



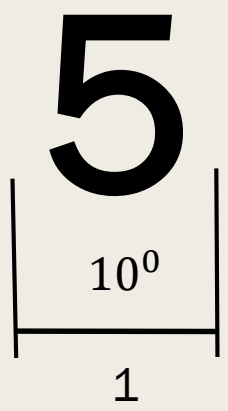
Base 10
Denary



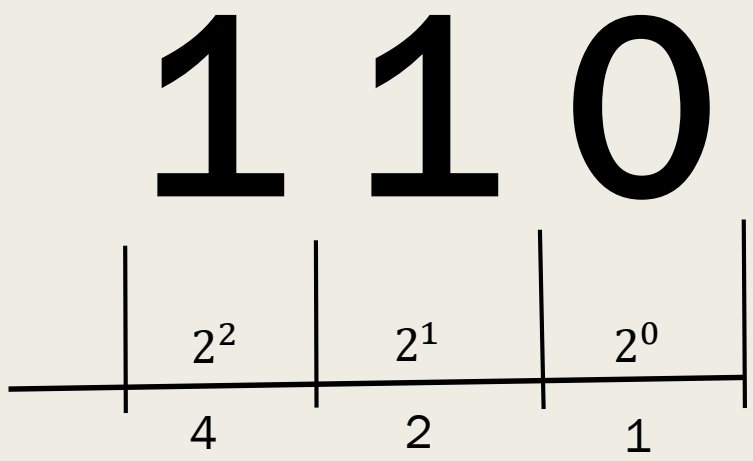
Base 2
Binary



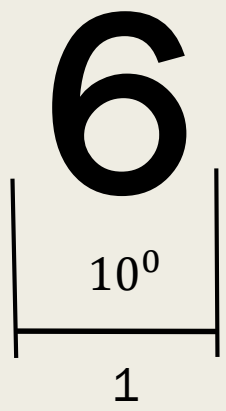
Base 10
Denary



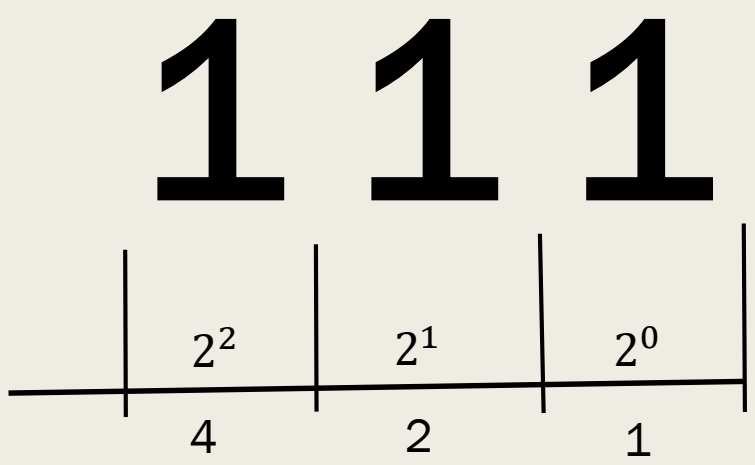
Base 2
Binary



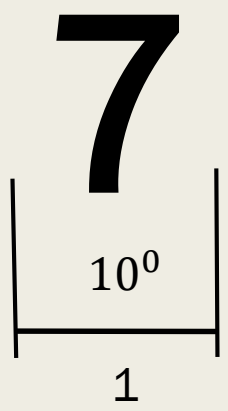
Base 10
Denary



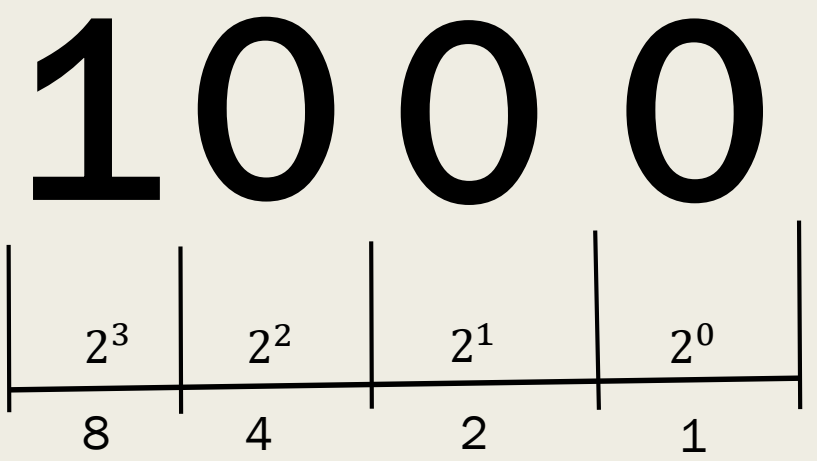
Base 2
Binary



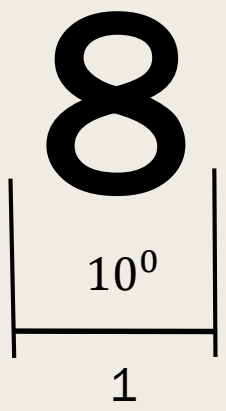
Base 10
Denary



Base 2
Binary



Base 10
Denary

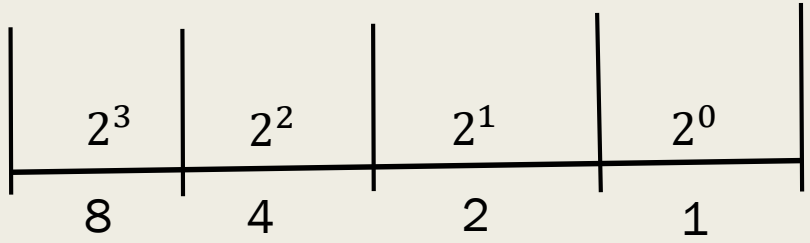


An exercise for you:

Convert the 4-bit **Binary** value **1100** into **Denary**:

Base 2
Binary

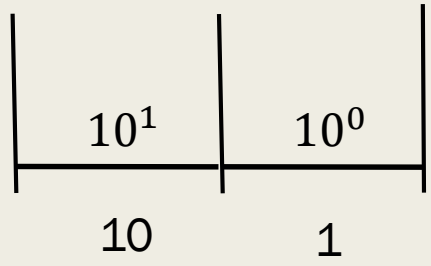
1 1 0 0



Base 10
Denary

8 + 4 = 12

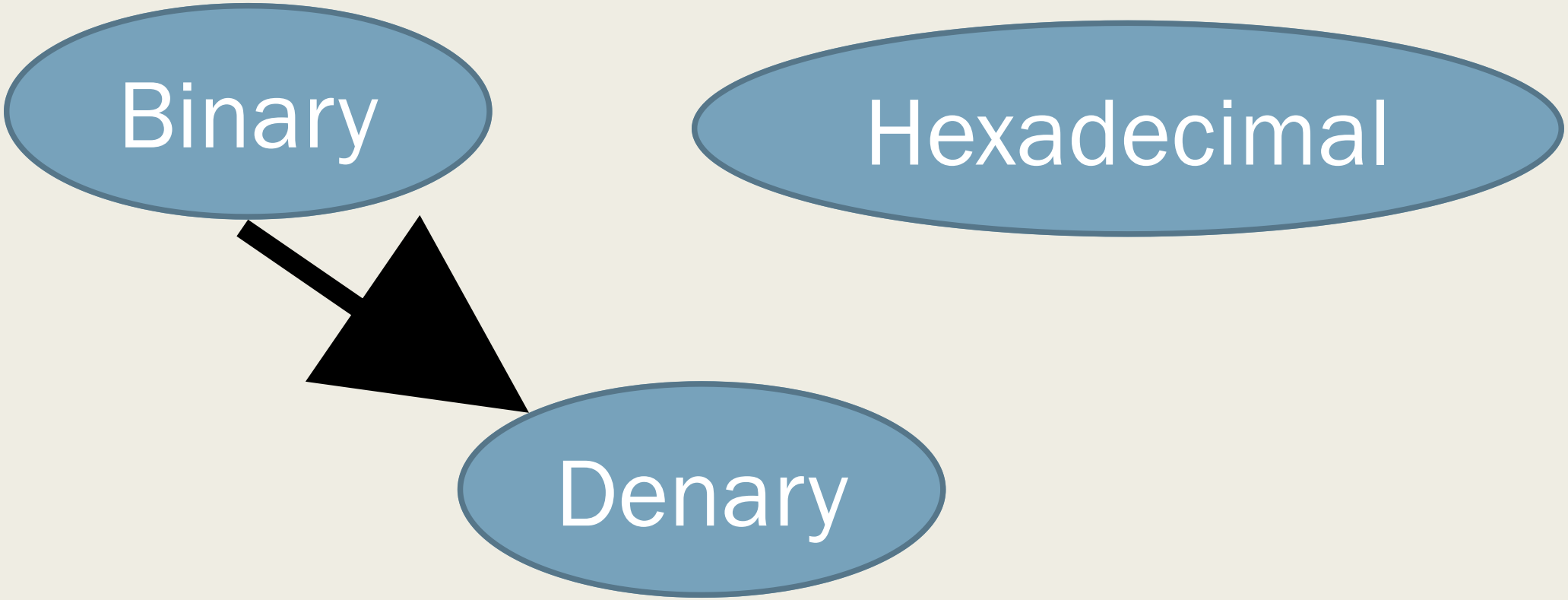
1 2



Tutorial exercise:

- This will provide you with more 4-bit Binary Nibbles to convert to Denary

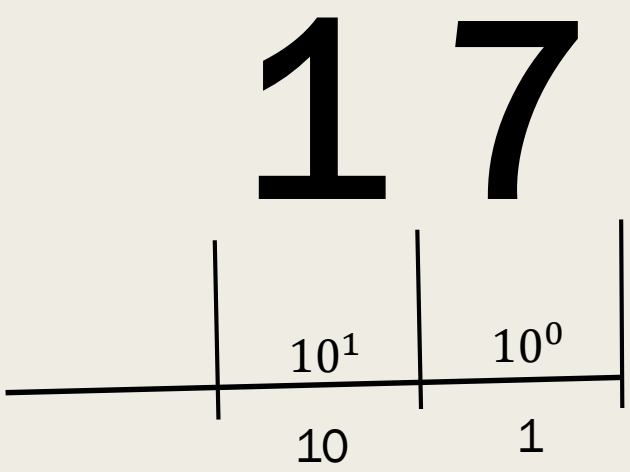
Number System Triangle



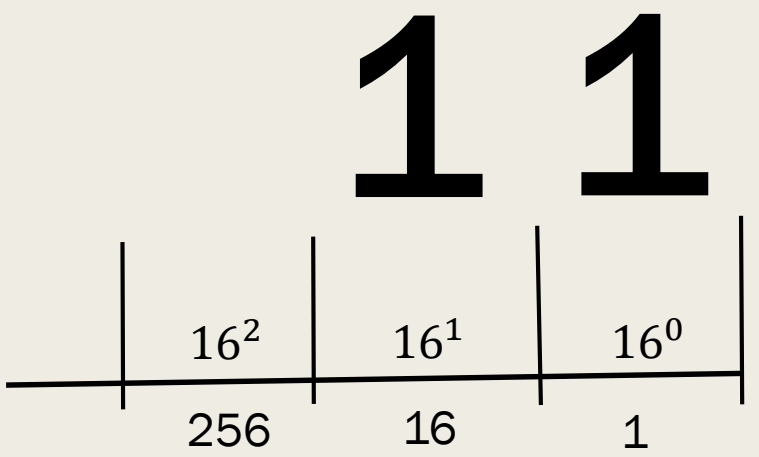
Hexadecimal table:

Denary	Binary	Hexadecimal
0	0 0 0 0	0
1	0 0 0 1	1
2	0 0 1 0	2
3	0 0 1 1	3
4	0 1 0 0	4
5	0 1 0 1	5
6	0 1 1 0	6
7	0 1 1 1	7
8	1 0 0 0	8
9	1 0 0 1	9
10	1 0 1 0	A
11	1 0 1 1	B
12	1 1 0 0	C
13	1 1 0 1	D
14	1 1 1 0	E
15	1 1 1 1	F

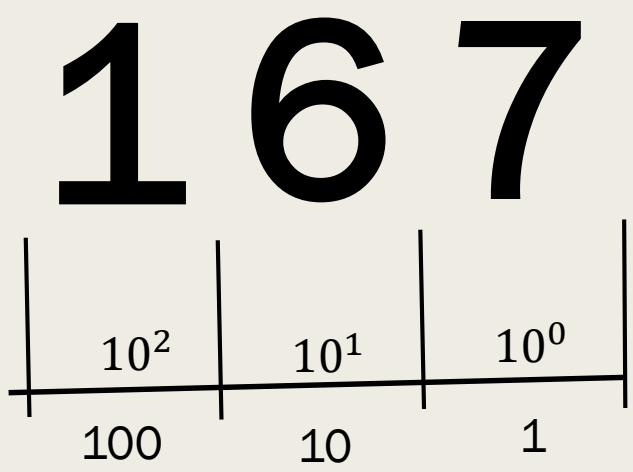
Base 10
Denary



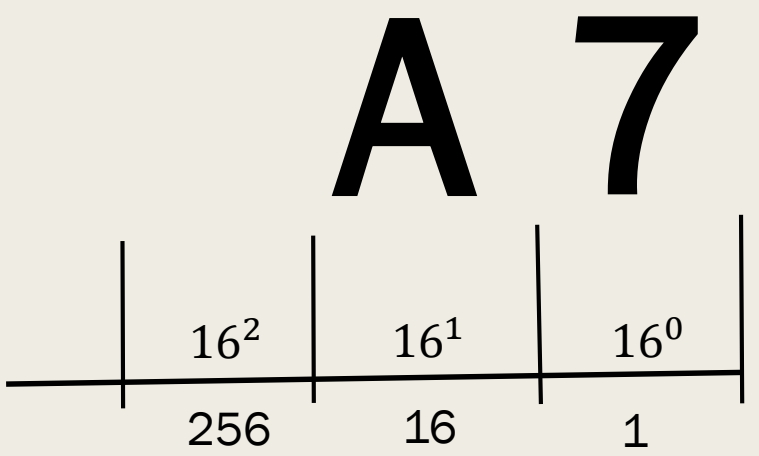
Base 16
Hexadecimal



Base 10
Denary



Base 16
Hexadecimal



In this lecture we have covered:

- Positional Number Systems
- Positive (unsigned) Integers
  - *Decimal / Denary – Base 10*
  - *Binary - Base 2*
  - *Hexadecimal – Base 16*

In the next lecture we will cover:

- Converting Denary to Binary

Thank you

© The University of Westminster (2021)

These notes were modified from the lecture slides generated by Noam Weingarten.
The right of Noam Weingarten to be identified as author of this work has been asserted by them in accordance with the Copyright, Designs and Patents Act 1988